

Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026

Topic 1

High Dimensional Probability and Statistics

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Chapter 1

Basic tail and concentration bounds

1.1 Introduction

A central challenge in high-dimensional statistics is the *curse of dimensionality*. As the dimension d grows, the sample space expands rapidly, so a fixed number of observations becomes increasingly sparse relative to the size of the space. As a result, methods and intuition that work well in low dimensions may become much less informative in high-dimensional settings.

High-dimensional problems also exhibit geometric and probabilistic behavior that can differ substantially from low-dimensional intuition. For example, if

$$X \sim N(0, I_d),$$

then the density is maximized at the origin, but a typical sample is not concentrated near the origin. Instead,

$$\|X\|_2 \approx \sqrt{d},$$

so most samples lie near a thin shell of radius approximately $\sqrt{d}$.

These phenomena motivate the study of probabilistic tools that remain useful when the dimension is large. In particular, this lecture develops tail and concentration bounds that quantify how likely a random variable is to deviate from its mean, with an emphasis on non-asymptotic guarantees.

1.1.1 One-dimensional and multivariate CLT

Let $X_1, \dots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2 . The one-dimensional CLT states that

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) \xrightarrow{d} N(0, \sigma^2) \quad \text{as } n \rightarrow \infty. \quad (1.1)$$

The multivariate version has the same form. Suppose now that $X_1, \dots, X_n$ are i.i.d. random vectors in $\mathbb{R}^d$ with

$$\mathbb{E}[X_i] = \mu \in \mathbb{R}^d, \quad \text{Cov}(X_i) = \Sigma.$$

Then

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) \xrightarrow{d} N(0, \Sigma) \quad \text{as } n \rightarrow \infty. \quad (1.2)$$

For a d -dimensional Gaussian vector $X \sim N(\mu, \Sigma)$ with positive definite covariance matrix Σ , the density is

$$f(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right\}. \quad (1.3)$$

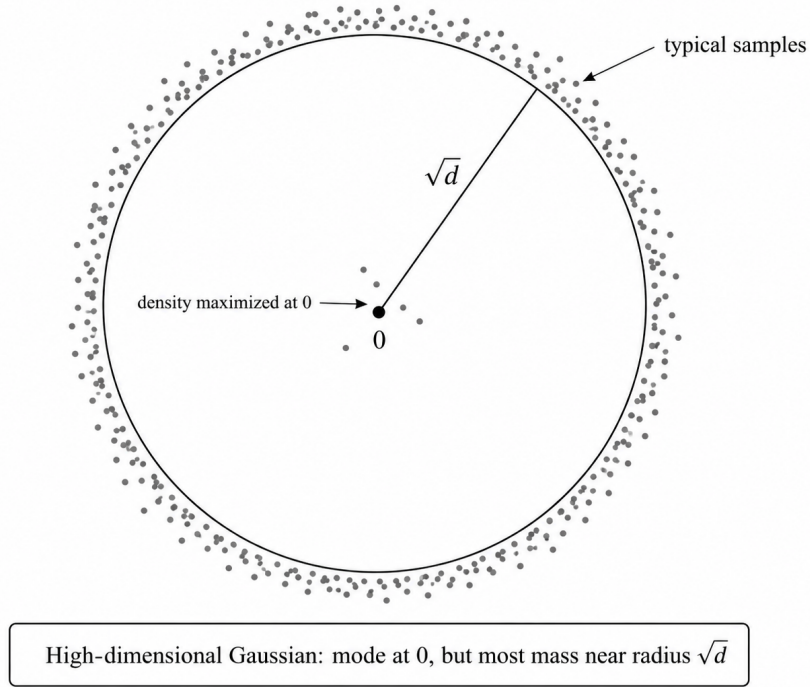


Figure 1.1: A schematic illustration of a high-dimensional standard Gaussian distribution.

Example 1.1.1 (A First High-Dimensional Phenomenon). Consider the standard Gaussian case $X \sim N(0, I_d)$. The density is maximized at the origin, so the mode is $x = 0$. However, when d is large, a typical draw is not close to the origin. Indeed,

$$\mathbb{E}\|X\|_2^2 = \mathbb{E}\left[\sum_{j=1}^d X_j^2\right] = \sum_{j=1}^d \mathbb{E}[X_j^2] = d. \quad (1.4)$$

Thus the typical Euclidean radius is of order $\sqrt{d}$; more precisely, $\|X\|_2^2$ is a chi-square random variable with d degrees of freedom and is concentrated around d . Hence most of the Gaussian probability mass lies in a thin shell with radius approximately $\sqrt{d}$, even though the density itself is largest at the origin, shown in Figure 1. This illustrates a basic feature of high-dimensional geometry: the location of the mode need not describe where a typical sample lies.

1.2 Classical Tail Bounds

The purpose of a tail bound is to control probabilities such as $\mathbb{P}(X \geq t)$ or $\mathbb{P}(|X - \mathbb{E}X| \geq t)$. The amount of information available about the distribution determines the quality of the bound. Markov's inequality uses only a first moment; Chebyshev's inequality uses a second moment; and the Chernoff method exploits the moment generating function (MGF), thereby incorporating information from all moments when the MGF exists [1].

1.2.1 Markov and Chebyshev inequalities

Proposition 1.2.1 (Markov's inequality). *If random variable X is nonnegative and $\mathbb{E}[X] < \infty$, then for*

every $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}. \quad (1.5)$$

Proposition 1.2.2 (Chebyshev's inequality). *If random variable X has finite variance and $\mu = \mathbb{E}[X]$, then for every $t > 0$,*

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\text{Var}(X)}{t^2}. \quad (1.6)$$

Chebyshev's inequality is an immediate consequence of Markov's inequality applied to the nonnegative random variable $(X - \mu)^2$. More generally, if the k th centered absolute moment exists, then

$$\mathbb{P}(|X - \mu| \geq t) = \mathbb{P}(|X - \mu|^k \geq t^k) \leq \frac{\mathbb{E}|X - \mu|^k}{t^k}. \quad (1.7)$$

1.2.2 From Markov to the Chernoff bound

The exponential function is particularly effective because it converts sums into products and leads naturally to the moment generating function.

Lemma 1.2.3 (Chernoff bound). *Let $\mu = \mathbb{E}[X]$, and suppose*

$$\phi(\lambda) = \mathbb{E}[e^{\lambda(X-\mu)}]$$

exists for $\lambda \in [0, b]$. Then for every $t > 0$,

$$\log \mathbb{P}(X - \mu \geq t) \leq \inf_{\lambda \in [0, b]} \{\log \phi(\lambda) - \lambda t\}. \quad (1.8)$$

Proof. For $\lambda > 0$. Since the exponential function is increasing,

$$\mathbb{P}(X - \mu \geq t) = \mathbb{P}(\lambda(X - \mu) \geq \lambda t) \quad (1.9)$$

$$= \mathbb{P}(e^{\lambda(X-\mu)} \geq e^{\lambda t}). \quad (1.10)$$

Applying Markov's inequality to the nonnegative random variable $e^{\lambda(X-\mu)}$ gives

$$\mathbb{P}(X - \mu \geq t) \leq \frac{\mathbb{E}[e^{\lambda(X-\mu)}]}{e^{\lambda t}} = \phi(\lambda)e^{-\lambda t}. \quad (1.11)$$

Taking logarithms yields

$$\log \mathbb{P}(X - \mu \geq t) \leq \log \phi(\lambda) - \lambda t.$$

Because the inequality holds for every admissible λ , minimizing the right-hand side over $\lambda \in [0, b]$ gives the result. $\square$

Remark 1.2.4. The optimization over λ is essential. Different values of λ produce different valid bounds, and the infimum selects the sharpest one obtained from this argument. In applications, the main work is therefore to control the MGF and then solve a one-dimensional optimization problem.

1.3 Sub-Gaussian Random Variables

1.3.1 Gaussian Tail Bound

As a first example, consider a Gaussian random variable

$$X \sim N(\mu, \sigma^2).$$

Its moment generating function is

$$\mathbb{E}[e^{\lambda X}] = \exp\left(\mu\lambda + \frac{\sigma^2\lambda^2}{2}\right), \quad \lambda \in \mathbb{R}. \quad (1.12)$$

Applying the preceding lemma, the upper-tail bound is determined by

$$\inf_{\lambda>0} \left\{ \log \mathbb{E}[e^{\lambda(X-\mu)}] - \lambda t \right\} = \inf_{\lambda>0} \left\{ \frac{\sigma^2\lambda^2}{2} - \lambda t \right\}. \quad (1.13)$$

Here we used the fact that

$$X - \mu \sim N(0, \sigma^2).$$

The quadratic expression is minimized at

$$\lambda^* = \frac{t}{\sigma^2},$$

which gives

$$\inf_{\lambda>0} \left\{ \frac{\sigma^2\lambda^2}{2} - \lambda t \right\} = -\frac{t^2}{2\sigma^2}. \quad (1.14)$$

Therefore, for every $t > 0$,

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right). \quad (1.15)$$

(Normal distribution is light tail)

1.3.2 Definition and basic tail bounds

Definition 1.3.1 (Sub-Gaussian random variable). A random variable X with mean $\mu = \mathbb{E}[X]$ is *sub-Gaussian with parameter $\sigma > 0$* if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right) \quad \text{for all } \lambda \in \mathbb{R}. \quad (1.16)$$

Remark 1.3.2. If X is σ -sub-Gaussian for some $\sigma > 0$, then X is also σ' -sub-Gaussian for some $\sigma' \geq \sigma$.

The definition says that the centered MGF of X grows no faster than the centered MGF of a Gaussian $N(0, \sigma^2)$. Therefore, the same Chernoff calculation gives the Gaussian-type upper-tail bound

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right), \quad t > 0. \quad (1.17)$$

Lemma 1.3.3 (Symmetry of sub-Gaussianity). *A random variable X is sub-Gaussian with parameter σ if and only if $-X$ is sub-Gaussian with the same parameter.*

Proof. Let $\mu = \mathbb{E}[X]$. Recall that X is sub-Gaussian with parameter σ if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right), \quad \lambda \in \mathbb{R}. \quad (1.18)$$

Suppose first that X is sub-Gaussian. Since $\mathbb{E}[-X] = -\mu$,

$$\mathbb{E}\left[e^{\lambda((-X)-\mathbb{E}[-X])}\right] = \mathbb{E}\left[e^{-\lambda(X-\mu)}\right] \quad (1.19)$$

$$\leq \exp\left(\frac{\sigma^2(-\lambda)^2}{2}\right) \quad (1.20)$$

$$= \exp\left(\frac{\sigma^2\lambda^2}{2}\right). \quad (1.21)$$

Hence $-X$ is sub-Gaussian with parameter σ .

Conversely, suppose that $-X$ is sub-Gaussian. Then

$$\mathbb{E} \left[e^{\lambda(X-\mu)} \right] = \mathbb{E} \left[e^{-\lambda((-X)-\mathbb{E}[-X])} \right] \quad (1.22)$$

$$\leq \exp \left(\frac{\sigma^2(-\lambda)^2}{2} \right) \quad (1.23)$$

$$= \exp \left(\frac{\sigma^2\lambda^2}{2} \right). \quad (1.24)$$

Therefore X is sub-Gaussian with parameter σ . $\square$

Lemma 1.3.4 (Lower Tail). *If X is sub-Gaussian with parameter σ , then for every $t > 0$,*

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) \leq \exp \left(-\frac{t^2}{2\sigma^2} \right). \quad (1.25)$$

Proof. By the symmetry lemma, $-X$ is also sub-Gaussian with parameter σ . Moreover,

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) = \mathbb{P}(-X - \mathbb{E}[-X] \geq t) \quad (1.26)$$

$$\leq \exp \left(-\frac{t^2}{2\sigma^2} \right), \quad (1.27)$$

where the last step follows from the upper-tail bound applied to $-X$. $\square$

Now if we combine upper and lower tails with the union bound yields the two-sided concentration inequality we will get the following

Lemma 1.3.5. *If X is sub-Gaussian with parameter $\sigma > 0$,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq 2 \exp \left(-\frac{t^2}{2\sigma^2} \right), \quad t > 0. \quad (1.28)$$

Note: This much faster decay is one reason sub-Gaussian assumptions are central in non-asymptotic high-dimensional statistics [1].

Example 1.3.6 (Rademacher random variable). A Rademacher random variable ε takes values in $\{-1, +1\}$ equiprobably:

In particular, $\mathbb{E}[\varepsilon] = 0$. Its MGF is

$$\mathbb{E}[e^{\lambda\varepsilon}] = \frac{1}{2}e^\lambda + \frac{1}{2}e^{-\lambda} \quad (1.29)$$

$$= \frac{1}{2} \left(\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} + \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!} \right) \quad (1.30)$$

$$= \sum_{m=0}^{\infty} \frac{\lambda^{2m}}{(2m)!}, \quad (1.31)$$

where the odd-order terms cancel.

Since $(2m)! \geq 2^m m!$,

$$\mathbb{E}[e^{\lambda\varepsilon}] \leq \sum_{m=0}^{\infty} \frac{\lambda^{2m}}{2^m m!} \quad (1.32)$$

$$= e^{\lambda^2/2}. \quad (1.33)$$

Since $\mathbb{E}[\varepsilon] = 0$, it follows that

$$\mathbb{E}[e^{\lambda(\varepsilon - \mathbb{E}[\varepsilon])}] \leq e^{\lambda^2/2}.$$

Therefore, ε is sub-Gaussian with parameter $\sigma = 1$.

This example shows that sub-Gaussianity is not the same as Gaussianity. A Rademacher variable is discrete and bounded, yet its MGF is dominated by that of a standard Gaussian. The sub-Gaussian class is therefore broad enough to include many non-Gaussian random variables while retaining Gaussian-like tail behavior.

Note: "equiprobably" meaning

$$\mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2}. \quad (1.34)$$

Therefore,

$$\mathbb{E}[\varepsilon] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0. \quad (1.35)$$

1.3.3 Example of Sub-Gaussian Variables

Example 1.3.7 (Bounded random variables). Let X be zero-mean, and supported on some interval $[a, b]$. We will show that X is sub-Gaussian with parameter at most $\sigma = b - a$.

Letting X' be an independent copy of X , so that $X \stackrel{d}{=} X'$ and $X \perp\!\!\!\perp X'$. For any $\lambda \in \mathbb{R}$, we have

$$\mathbb{E}_X [e^{\lambda X}] = \mathbb{E}_X [e^{\lambda(X - \mathbb{E}_{X'}[X'])}] \stackrel{(i)}{\leq} \mathbb{E}_X [\mathbb{E}_{X'} e^{\lambda(X - X')}] = \mathbb{E}_{X, X'} [e^{\lambda(X - X')}]$$

The first equality is because $\mathbb{E}[X'] = 0$, and (i) follows from the the convexity of the exponential, and Jensen's inequality. Letting ε be an independent Rademacher variable, note that since (X, X') are i.i.d, $X - X' \stackrel{d}{=} X' - X$, so $(X - X')$ is symmetric about 0. Thus, the distribution of $(X - X')$ is the same as that of $\varepsilon(X - X')$, so that we have

$$\mathbb{E}_{X, X'} [e^{\lambda(X - X')}] = \mathbb{E}_{X, X'} [\mathbb{E}_\varepsilon [e^{\lambda\varepsilon(X - X')}]] \stackrel{(ii)}{\leq} \mathbb{E}_{X, X'} [e^{\frac{\lambda^2}{2}(X - X')^2}] \leq e^{\frac{\lambda^2}{2}(b-a)^2}$$

The first equality follows from Tonelli's theorem and (ii) follows from the sub-Gaussianity of Radamacher variable and last inequality is because $|X - X'| \leq b - a$. Putting everything together, we have shown that X is sub-Gaussian with parameter at most $\sigma = b - a$.

Remark 1.3.8. This result is useful, but it can be sharpened. It can be shown that X is sub-Gaussian with parameter at most $\sigma = \frac{b-a}{2}$ (Exercise 2.4). The technique used in this example is a simple example of a *symmetrization argument*, in which we first introduce an independent copy X' , and then symmetrize the problem with a Rademacher variable. Such symmetrization arguments are useful in a variety of contexts.

1.3.4 Hoeffding's Inequality

Another property of sub-Gaussianity is preserved by linear operations.

Lemma 1.3.9. *Let X_1 and X_2 be independent sub-Gaussian variables with parameters σ_1 and σ_2 , respectively, then $X_1 + X_2$ is sub-Gaussian with parameter $\sqrt{\sigma_1^2 + \sigma_2^2}$.*

Proof. By independence we have

$$\mathbb{E} [e^{\lambda(X_1 + X_2 - \mathbb{E}X_1 - \mathbb{E}X_2)}] = \mathbb{E} [e^{\lambda(X_1 - \mathbb{E}X_1)}] \mathbb{E} [e^{\lambda(X_2 - \mathbb{E}X_2)}] \leq e^{\frac{\lambda^2}{2}\sigma_1^2} e^{\frac{\lambda^2}{2}\sigma_2^2} = e^{\frac{\lambda^2}{2}(\sigma_1^2 + \sigma_2^2)}$$

□

As a consequence of this fact, we obtain an important result known as the *Hoeffding's inequality*:

Proposition 1.3.10 (Hoeffding's inequality). *Suppose that the variables X_i , $i = 1, \dots, n$, are independent, and X_i has mean μ_i and sub-Gaussian parameter σ_i . Then for all $t \geq 0$, we have*

$$\mathbb{P} \left[\sum_{i=1}^n (X_i - \mu_i) \geq t \right] \leq \exp \left\{ -\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2} \right\}$$

and

$$\mathbb{P} \left[\left| \sum_{i=1}^n X_i - \sum_{i=1}^n \mathbb{E} X_i \right| \geq t \right] \leq 2 \exp \left\{ -\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2} \right\}$$

Proof. By previous lemma, $\sum_{i=1}^n X_i$ is $\sqrt{\sigma_1^2 + \dots + \sigma_n^2}$ sub-Gaussian. So, by basic sub-Gaussian tail bound we have the results. $\square$

Corollary 1.3.11. *If X_i , $i = 1, \dots, n$, are i.i.d, with mean μ and sub-Gaussian parameter σ . Then for all $t \geq 0$, we have*

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq t \right] \leq 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$$

and

$$\mathbb{P} \left[\sqrt{n} \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq t \right] \leq 2 \exp \left\{ -\frac{t^2}{2\sigma^2} \right\}$$

Remark 1.3.12. The above can be viewed as a finite sample Central Limit Theorem. CLT states that when $n \rightarrow \infty$,

$$\sqrt{n} \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \rightarrow \mathcal{N}(0, \sigma^2).$$

The second statement of the above corollary is a finite sample version of CLT, by stating that the tail of $\sqrt{n} \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right|$ is upper bounded by the tail of $\mathcal{N}(0, \sigma^2)$.

Proof. By Hoeffding,

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq t \right] = \mathbb{P} \left[\left| \sum_{i=1}^n X_i - n\mu \right| \geq nt \right] \leq 2 \exp \left\{ -\frac{(nt)^2}{2n\sigma^2} \right\} = 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$$

Second statement follows analogously. $\square$

One application of Hoeffding's bound is to derive finite sample guarantees for the error of the sample mean.

Example 1.3.13 (Finite-sample error guarantee). Let $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$ and fix a confidence level $\delta \in (0, 1)$. By the preceding corollary,

$$\mathbb{P} [|\bar{X}_n - \mu| \geq t] \leq 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$$

Choose $t = \sqrt{\frac{2\sigma^2}{n} \log \left(\frac{2}{\delta} \right)}$, so that $\delta = 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$. In other words, with probability at least $1 - \delta$,

$$|\bar{X}_n - \mu| \leq \sqrt{\frac{2\sigma^2}{n} \log \left(\frac{2}{\delta} \right)}$$

Example 1.3.14 (Expected absolute error). We can also control the expected absolute error $\mathbb{E}|\bar{X}_n - \mu|$ directly from the tail bound. Recall that for any nonnegative random variable Y , Tonelli's theorem gives

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y \geq t) dt$$

Taking $Y = |\bar{X}_n - \mu|$, the preceding corollary and the trivial bound $\mathbb{P}(Y \geq t) \leq 1$ imply

$$\mathbb{P}(|\bar{X}_n - \mu| \geq t) \leq \min \left\{ 1, 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\} \right\}$$

Let t_0 be the point at which the two upper bounds agree. Thus $2e^{-\frac{nt_0^2}{2\sigma^2}} = 1$, which gives $t_0 = \sigma \sqrt{\frac{2 \log 2}{n}}$. Hence

$$\begin{aligned} \mathbb{E}|\bar{X}_n - \mu| &= \int_0^\infty \mathbb{P}(|\bar{X}_n - \mu| \geq t) dt \\ &\leq \int_0^{t_0} 1 dt + \underbrace{\int_{t_0}^\infty 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\} dt}_{z = \frac{\sqrt{n}}{\sigma} t, \quad dt = \frac{\sigma}{\sqrt{n}} dz} \\ &= t_0 + \frac{2\sigma}{\sqrt{n}} \int_{\frac{\sqrt{n}}{\sigma} t_0}^\infty e^{-z^2/2} dz \\ &= \frac{\sigma}{\sqrt{n}} \left(\sqrt{2 \log 2} + 2 \int_{\sqrt{2 \log 2}}^\infty e^{-z^2/2} dz \right) \\ &\leq 2 \frac{\sigma}{\sqrt{n}}. \end{aligned}$$

We conclude our discussion of sub-Gaussianity with a theorem that provides different characterizations of sub-Gaussian variables.

Theorem 1.3.15 (Equivalent characterizations of sub-Gaussian variables). *Given any zero-mean random variable X , the following properties are equivalent:*

1. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E}[e^{\lambda X}] \leq e^{\frac{\lambda^2 \sigma^2}{2}} \quad \text{for all } \lambda \in \mathbb{R}.$$

2. *There is a constant $c \geq 0$ and Gaussian random variable $Z \sim \mathcal{N}(0, \tau^2)$ such that*

$$\mathbb{P}(|X| \geq s) \leq c \mathbb{P}(|Z| \geq s) \quad \text{for all } s \geq 0.$$

3. *There is a constant $\theta \geq 0$ such that*

$$\mathbb{E}[X^{2k}] \leq \frac{(2k)!}{2^k k!} \theta^{2k} \quad \text{for all } k = 1, 2, \dots$$

4. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E} \left[e^{\frac{\lambda X^2}{2\sigma^2}} \right] \leq \frac{1}{\sqrt{1-\lambda}} \quad \text{for all } \lambda \in [0, 1).$$

Proof. See Appendix A (Section 2.4) in [1]. □

1.4 Symmetrization

Lemma 1.4.1. *Let $Y \perp\!\!\!\perp Z$ with $\mathbb{E}[Z] = 0$. Then for any convex function F ,*

$$\mathbb{E}[F(Y)] \leq \mathbb{E}[F(Y + Z)]. \quad (1.36)$$

Example 1.4.2. The function $F(x) = e^{\lambda x}$ is convex. Suppose $\mathbb{E}[X] = 0$. Applying Lemma 1.4.1 with $Y = X$ and $Z = -X'$, where $X' \perp\!\!\!\perp X$ and $X' \sim \text{Law}(X)$, we obtain

$$\mathbb{E}_X [e^{\lambda X}] \leq \mathbb{E}_{X, X'} [e^{\lambda(X - X')}] \quad (1.37)$$

Proof of Lemma 1.4.1. For every $y \in \mathbb{R}$, using $\mathbb{E}[Z] = 0$ and Jensen's inequality,

$$F(y) = F(y + \mathbb{E}_Z[Z]) = F(\mathbb{E}_Z[y + Z]) \leq \mathbb{E}_Z[F(y + Z)].$$

Substituting $y = Y$ gives $F(Y) \leq \mathbb{E}_Z[F(Y + Z)]$, and taking expectations over Y yields

$$\mathbb{E}_Y[F(Y)] \leq \mathbb{E}_Y \mathbb{E}_Z[F(Y + Z)] = \mathbb{E}_{Y, Z}[F(Y + Z)],$$

where the last equality uses $Y \perp\!\!\!\perp Z$. □

Definition 1.4.3. A random variable X is called *symmetric* if it has the same distribution as $-X$.

Remark 1.4.4. If X has pdf $f(x)$, then X is symmetric if and only if $f(x) = f(-x)$.

Lemma 1.4.5 (Constructing symmetric distributions). *Let X be a random variable and let ε be an independent Rademacher random variable.*

- (a) εX and $\varepsilon|X|$ are identically distributed and symmetric.
- (b) If X is symmetric, then both εX and $\varepsilon|X|$ have the same distribution as X .
- (c) If X' is an independent copy of X , then $X - X'$ is symmetric.

Proof. We only prove that εX is symmetric. For every event $A \subseteq \mathbb{R}$, conditioning on the value of ε gives

$$\mathbb{P}[\varepsilon X \in A] = \frac{1}{2}\mathbb{P}[X \in A] + \frac{1}{2}\mathbb{P}[-X \in A] = \frac{1}{2}\mathbb{P}[-X \in A] + \frac{1}{2}\mathbb{P}[X \in A] = \mathbb{P}[-\varepsilon X \in A]. \quad \square$$

Lemma 1.4.6 (Symmetrization). *Let $X_1, \dots, X_N$ be independent, mean-zero random vectors in some normed space, and let $\varepsilon_1, \dots, \varepsilon_N$ be independent Rademacher random variables. Then*

$$\frac{1}{2} \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \leq 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|. \quad (1.38)$$

Proof. Let $(X'_1, \dots, X'_N)$ be independent copies of $(X_1, \dots, X_N)$. By Lemma 1.4.5(c) the vectors $X_i - X'_i$ are independent and symmetric (this step uses the assumption that X_i are independent), so that

$$(X_1 - X'_1, \dots, X_N - X'_N) \stackrel{d}{=} (\varepsilon_1(X_1 - X'_1), \dots, \varepsilon_N(X_N - X'_N)). \quad (1.39)$$

Upper bound. Set $p := \mathbb{E} \left\| \sum_{i=1}^N X_i \right\|$. Since $\mathbb{E} \left[\sum_{i=1}^N X'_i \right] = 0$, we may apply Lemma 1.4.1 with $F(x) = \|x\|$, $Y = \sum_{i=1}^N X_i$ and $Z = -\sum_{i=1}^N X'_i$ to get

$$p = \mathbb{E} \left\| \sum_{i=1}^N X_i - \mathbb{E} \left[\sum_{i=1}^N X'_i \right] \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N (X_i - X'_i) \right\|.$$

Then by (1.39) and the triangle inequality,

$$\begin{aligned} \mathbb{E} \left\| \sum_{i=1}^N (X_i - X'_i) \right\| &= \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i (X_i - X'_i) \right\| = \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i - \sum_{i=1}^N \varepsilon_i X'_i \right\| \\ &\leq \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| + \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X'_i \right\| = 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|. \end{aligned}$$

Lower bound. Conditionally on $\varepsilon = (\varepsilon_1, \dots, \varepsilon_N)$, the vectors $\sum_{i=1}^N \varepsilon_i X_i$ and $-\sum_{i=1}^N \varepsilon_i X'_i$ are independent and the latter has mean zero, so Lemma 1.4.1 applies conditionally and gives, after averaging over ε ,

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i - \sum_{i=1}^N \varepsilon_i X'_i \right\| = \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i (X_i - X'_i) \right\|.$$

Applying (1.39) once more and then the triangle inequality,

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i (X_i - X'_i) \right\| = \mathbb{E} \left\| \sum_{i=1}^N (X_i - X'_i) \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| + \mathbb{E} \left\| \sum_{i=1}^N X'_i \right\| = 2 \mathbb{E} \left\| \sum_{i=1}^N X_i \right\|. \quad \square$$

Remark 1.4.7. The zero-mean assumption is needed for both bounds: in the upper bound to subtract $\mathbb{E}[\sum_i X'_i]$, and in the lower bound to guarantee that $\mathbb{E}[\sum_i \varepsilon_i X'_i \mid \varepsilon] = 0$.

1.5 Sub-exponential Random Variables

We now turn to the class of sub-exponential variables, which are defined by a slightly milder condition on the moment generating function.

Definition 1.5.1 ([1, Definition 2.7]). A random variable X with mean $\mu = \mathbb{E}[X]$ is *sub-exponential* if there are non-negative parameters (ν, α) such that

$$\mathbb{E} \left[e^{\lambda(X-\mu)} \right] \leq e^{\frac{\nu^2 \lambda^2}{2}} \quad \text{for all } |\lambda| < \frac{1}{\alpha}. \quad (1.40)$$

Remark 1.5.2. A σ -sub-Gaussian variable is $(\sigma, 0)$ -sub-exponential, where we interpret $1/0$ as $+\infty$.

Example 1.5.3 (Sub-exponential but not sub-Gaussian). Let $Z \sim N(0, 1)$ and consider $X = Z^2$, so that $\mathbb{E}[X] = 1$. For $\lambda < \frac{1}{2}$,

$$\begin{aligned} \mathbb{E} \left[e^{\lambda(X-1)} \right] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{\lambda(z^2-1)} e^{-\frac{z^2}{2}} dz \\ &= \frac{e^{-\lambda}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{(1-2\lambda)z^2}{2}} dz = \frac{e^{-\lambda}}{\sqrt{1-2\lambda}}, \end{aligned} \quad (1.41)$$

where the last equality uses $\int_{-\infty}^{+\infty} e^{-az^2/2} dz = \sqrt{2\pi/a}$ with $a = 1 - 2\lambda > 0$. For $\lambda \geq \frac{1}{2}$ the moment generating function is infinite, so X is not sub-Gaussian. On the other hand,

$$\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{2\lambda^2} = e^{\frac{2^2 \lambda^2}{2}} \quad \text{where } |\lambda| < \frac{1}{4} \quad (\text{check this!}), \quad (1.42)$$

so that $\mathbb{E} \left[e^{\lambda(X-1)} \right] \leq e^{\frac{2^2 \lambda^2}{2}}$ for all $|\lambda| < \frac{1}{4}$, i.e. X is $(2, 4)$ -sub-exponential.

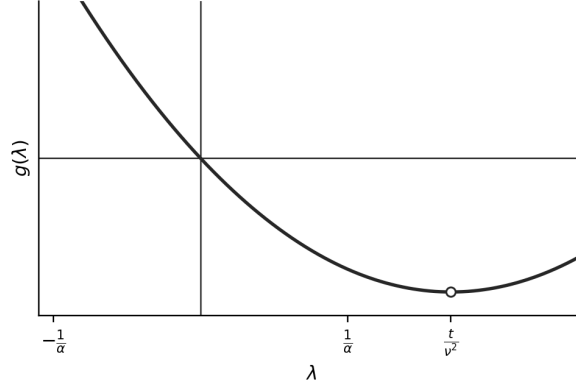


Figure 1.2: The function $g(\lambda) = \frac{\nu^2 \lambda^2}{2} - \lambda t$ appearing in (1.44), in the case $\frac{t}{\nu^2} > \frac{1}{\alpha}$. The open circle marks the unconstrained minimizer $\frac{t}{\nu^2}$, which lies outside the feasible region $|\lambda| < \frac{1}{\alpha}$.

Proposition 1.5.4 (Sub-exponential tail bound). *Suppose X is sub-exponential with parameters (ν, α) and mean μ . Then*

$$\mathbb{P}[X - \mu \geq t] \leq \begin{cases} e^{-\frac{t^2}{2\nu^2}}, & \text{if } 0 \leq t \leq \frac{\nu^2}{\alpha}, \\ e^{-\frac{t}{2\alpha}}, & \text{for } t > \frac{\nu^2}{\alpha}, \end{cases} = e^{-\min\left\{\frac{t^2}{2\nu^2}, \frac{t}{2\alpha}\right\}}. \quad (1.43)$$

Proof. Assume $\mu = 0$ without loss of generality. By the Chernoff bound from Lecture 1 and Definition 1.5.1,

$$\begin{aligned} \log \mathbb{P}[X \geq t] &\leq \inf_{0 \leq \lambda < \frac{1}{\alpha}} \{ \log \mathbb{E}[e^{\lambda X}] - \lambda t \} \leq \inf_{0 \leq \lambda < \frac{1}{\alpha}} \left\{ \log e^{\frac{\nu^2 \lambda^2}{2}} - \lambda t \right\} \\ &= \inf_{0 \leq \lambda < \frac{1}{\alpha}} \underbrace{\left\{ \frac{\nu^2 \lambda^2}{2} - \lambda t \right\}}_{=: g(\lambda)}. \end{aligned} \quad (1.44)$$

The function g is a convex parabola (see Figure 1.2) with

$$g'(\lambda) = \nu^2 \lambda - t = 0 \quad \Longleftrightarrow \quad \lambda = \frac{t}{\nu^2}.$$

① If $\frac{t}{\nu^2} < \frac{1}{\alpha}$ (i.e. $t < \frac{\nu^2}{\alpha}$), the unconstrained minimizer lies in the feasible region, so

$$\inf_{0 \leq \lambda < \frac{1}{\alpha}} g(\lambda) = g\left(\frac{t}{\nu^2}\right) = \frac{\nu^2}{2} \cdot \frac{t^2}{\nu^4} - \frac{t}{\nu^2} \cdot t = -\frac{t^2}{2\nu^2}.$$

② If $\frac{t}{\nu^2} \geq \frac{1}{\alpha}$ (i.e. $t \geq \frac{\nu^2}{\alpha}$), then g is decreasing on $[0, \frac{t}{\nu^2})$ and the infimum is approached at $\lambda = \frac{1}{\alpha}$, giving

$$\inf_{0 \leq \lambda < \frac{1}{\alpha}} g(\lambda) = g\left(\frac{1}{\alpha}\right) = \frac{\nu^2}{2} \cdot \frac{1}{\alpha^2} - \frac{1}{\alpha} t = \frac{\nu^2}{2\alpha^2} - \frac{t}{\alpha} \leq -\frac{t}{2\alpha},$$

where the last step uses $\nu^2 < \alpha t$. □

1.5.1 Bernstein's Condition

We say that X satisfies *Bernstein's condition with parameter b* if

$$|\mathbb{E}[(X - \mu)^k]| \leq \frac{1}{2} k! \operatorname{Var}(X) b^{k-2}, \quad \text{for any } k = 2, 3, \dots, \quad (1.45)$$

where $\mu = \mathbb{E}[X]$.

Example 1.5.5. If (1.45) is satisfied, then X is sub-exponential. Indeed, writing $\sigma^2 = \operatorname{Var}(X)$ and using the Taylor expansion of the exponential together with $\mathbb{E}[X - \mu] = 0$,

$$\begin{aligned} \mathbb{E}[e^{\lambda(X-\mu)}] &= \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{(\lambda(X-\mu))^k}{k!}\right] = 1 + 0 + \mathbb{E}\left[\sum_{k=2}^{\infty} \frac{\lambda^k (X-\mu)^k}{k!}\right] \\ &= 1 + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}[(X-\mu)^k]}{k!} = 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{\lambda^k \mathbb{E}[(X-\mu)^k]}{k!} \\ &\leq 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{|\lambda|^k \cdot \frac{1}{2} k! \sigma^2 b^{k-2}}{k!} = 1 + \frac{\lambda^2 \sigma^2}{2} + \frac{\lambda^2 \sigma^2}{2} \sum_{k=3}^{\infty} (|\lambda| b)^{k-2} = 1 + \frac{\lambda^2 \sigma^2}{2} \sum_{s=0}^{\infty} (|\lambda| b)^s. \end{aligned} \quad (1.46)$$

For $|\lambda| < \frac{1}{b}$ the geometric series sums to $\frac{1}{1-|\lambda|b}$, so that

$$\mathbb{E}[e^{\lambda(X-\mu)}] \leq 1 + \frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|} \leq e^{\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}}, \quad (1.47)$$

using $1 + t \leq e^t$. Restricting further to $|\lambda| < \frac{1}{2b}$ gives $1 - b|\lambda| > \frac{1}{2}$, and therefore

$$\mathbb{E}[e^{\lambda(X-\mu)}] \leq e^{\lambda^2 \sigma^2} = e^{\frac{\lambda^2 (\sqrt{2}\sigma)^2}{2}}, \quad \text{for all } |\lambda| < \frac{1}{2b},$$

i.e. X is sub-exponential with parameters $(\nu, \alpha) = (\sqrt{2}\sigma, 2b)$.

Exercises

1. Verify the inequality (1.42), that is, show that $\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{2\lambda^2}$ for all $|\lambda| < \frac{1}{4}$.

Bibliography

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